

Proposal for Neo4j Collaboration with Dinis Cruz

Executive Summary

Dinis Cruz is a seasoned cybersecurity expert and innovator with a long-standing relationship with Neo4j, marked by years of advocacy for graph technologies. He brings over two decades of experience in security and software engineering ¹ and has been **deeply passionate about graph databases** throughout his career. Notably, Dinis often quips that “everything is really just graphs and maps,” underscoring his conviction in the power of graph-based thinking ². His history with Neo4j dates back to pioneering projects where he leveraged Neo4j in novel ways (e.g. mapping GDPR data flows in 2018) and extends through ongoing dialogues on graph innovation.

Current Innovations: Dinis’s latest work centers on **Serverless Semantic Knowledge Graphs** – an approach to building knowledge graphs that harness serverless computing and AI. He recently open-sourced **MGraph-AI**, a memory-first graph database tailored for AI and **serverless** environments ³. Using this and other tools, Dinis has developed platforms like **MyFeeds.ai** (an LLM-driven semantic news feed) and is architecting **The Cyber Boardroom**, a GenAI-powered cybersecurity insights platform for corporate boards. These initiatives demonstrate how **graph databases plus AI** can deliver scalable, **deterministic** and **provenance-rich** insights in real-world applications ⁴ ⁵.

Past Neo4j Engagement: Dinis’s relationship with Neo4j is both practical and collaborative. As early as 2018, his team’s use of Neo4j to visualize complex security data was featured by Neo4j’s community ⁶. He has shared **ideas with Neo4j personnel** in venues like the Open Security Summit 2022, where he co-led a session (with Neo4j’s own experts in attendance) on “Ideas for Graph DBs like Neo4j” ⁷ ⁸. Over the years he has consistently championed graph database use in cybersecurity – from a 2017 DevSecCon keynote on graph-based security management, to OWASP and Open Security Summit talks – positioning Neo4j as a key technology in those discussions.

Opportunity for Neo4j: This proposal outlines strategic collaboration opportunities aligning Dinis’s innovative research with Neo4j’s technology direction. Areas of synergy include integrating **serverless graph** capabilities, co-developing solutions for **knowledge graphs + LLMs**, connectors for enterprise data (like Jira) into Neo4j, and joint thought leadership in cybersecurity use cases. By partnering with Dinis, Neo4j can leverage his open-source work and domain expertise to expand into new use-cases (e.g. board-level cyber risk knowledge graphs) and reinforce Neo4j’s position as the graph platform of choice for AI-driven applications. The following sections provide a detailed timeline of Dinis’s interactions with Neo4j, an overview of his current projects, and concrete proposals for collaboration – all with a focus on clear strategic alignment to Neo4j’s interests.

Dinis Cruz: Background and Graph Expertise

Dinis Cruz is a cybersecurity leader and technologist known for blending deep security expertise with software engineering skill ¹. Over a 20+ year career, he has served in roles such as CISO and CTO (e.g. at Glasswall) and has earned industry recognition (nominated **CISO of the Year 2019**, and ranked among the “10 Best CSOs of 2020” ⁹). His technical philosophy emphasizes enabling business through security innovation, with a focus on productizing advanced technologies.

Passion for Graph Technology: Among Dinis's core technical passions is graph data modeling. He has persistently advocated for the use of graph databases to solve complex security and knowledge management problems. He often emphasizes that many challenges in security and knowledge representation naturally fit graph structures – echoing his personal mantra that *“everything is really just graphs and maps”* ². This perspective has driven him to experiment with graph solutions in various domains (from threat modeling to compliance to AI knowledge bases), frequently positioning Neo4j as a preferred tool.

Dinis is also an active open-source contributor and thought leader. He has authored books and numerous articles, and spearheaded projects at the intersection of security and engineering. His background with OWASP (leading projects like the O2 Platform) and community initiatives (e.g. founding the Open Security Summit series) shows a commitment to collaborative innovation. In summary, Dinis brings a unique combination of **security domain knowledge**, **graph database expertise**, and **entrepreneurial drive** – making him well-suited to collaborate with Neo4j on forward-looking R&D initiatives.

Timeline of Engagements with Neo4j

To illustrate Dinis Cruz's long-standing relationship with Neo4j, below is a timeline highlighting key interactions, projects, and public talks that involved Neo4j:

Year	Engagement / Project	Details & Significance
2017	DevSecCon Keynote: Graph-Based Security	Dinis delivered a keynote “Creating a Graph-Based Security Organisation,” introducing how graph databases can model security operations and relationships (based on his work at Photobox). This early advocacy for graphs in security set the tone for future Neo4j applications ¹⁰ .
2018	Photobox GDPR Data Graph (Neo4j)	As CISO at Photobox, Dinis's team utilized Neo4j to map GDPR data journeys (personal data flows). A Neo4j + VisJS visualization of this project garnered attention and was featured as “Tweet of the Week” by Neo4j's community blog ⁶ , highlighting Dinis's innovative use of Neo4j in compliance/security.
2019	OWASP London Talk on Graph Security	Dinis presented his graph-based security approach to the OWASP community, reinforcing the concept of using graph databases (like Neo4j) to model security risks, identity relationships, and threat intel. This talk (an evolution of the 2017 keynote) underscored his continued commitment to graph solutions in cybersecurity.
2021	Exploration of Jira as a Graph DB	Through blogging and research, Dinis explored treating Atlassian Jira as a graph database (using issues as nodes/edges). He noted overlaps and gaps compared to Neo4j, identifying a “missed opportunity” in how little Jira had evolved as a graph store ¹¹ . This exploration informed his thinking on lightweight, flexible graph data storage and the potential advantages of Neo4j for certain use cases.

Year	Engagement / Project	Details & Significance
Mar 2022	Open Security Summit – “Ideas for Graph DBs like Neo4j”	Dinis co-led a public session focused on innovative graph database use cases with a special focus on Neo4j ¹² . Joined by Neo4j representatives and community experts, he discussed topics ranging from using graphs in threat modeling and risk management to “ Lightweight and Serverless Graph DBs ” and treating Neo4j instances as ephemeral resources ⁷ ⁸ . This session fostered collaborative brainstorming on how Neo4j could be applied in cybersecurity and DevOps, strengthening Dinis’s direct dialogue with Neo4j’s community.
2023	Development of The Cyber Boardroom (Prototype)	Dinis began building <i>The Cyber Boardroom</i> platform – a use case that combines graph knowledge bases with GenAI to help corporate boards manage cyber risks. Neo4j featured prominently in conceptual discussions as the kind of graph backend ideal for mapping complex relationships (threats, controls, business assets). Early prototypes used Dinis’s own graph tools, laying groundwork for later integration with Neo4j technology.
2024	Serverless Graph DB R&D (MGraph-AI)	After iterating on graph tools, Dinis released MGraph-AI in early 2025 (development in late 2024), an open-source Python graph database optimized for speed and serverless deployment ³ . This R&D was partly motivated by needs identified while using Neo4j in cloud environments – e.g. the desire for a graph DB with zero-cost idle time and instant spin-up . The work is complementary to Neo4j, and insights from it present an opportunity for Neo4j (e.g. in supporting ephemeral cloud functions) as discussed later in this proposal.
2025	MyFeeds.ai Launch (Semantic Knowledge Graphs + LLM)	Dinis launched <i>MyFeeds.ai</i> as a demonstration of Serverless Semantic Knowledge Graphs in action. The system uses LLMs to convert news articles into structured knowledge graphs , providing personalized cyber security news feeds with full provenance of why each item was selected ⁴ ⁵ . This project showcases cutting-edge integration of Neo4j-like graph concepts with AI – a clear intersection of Dinis’s work and Neo4j’s interests in knowledge graphs for AI.
2025	Cyber Boardroom Business Plan & Outreach	Dinis formalized <i>The Cyber Boardroom</i> vision in a business plan, highlighting it as a GenAI-powered, graph-backed platform to bridge boards and cybersecurity teams ¹³ . He is currently seeking seed investment (£250k for 20% equity) to accelerate this platform ¹⁴ . The plan underscores strategic opportunities (and potential partnership openings) for Neo4j – since the platform’s core involves semantic graph knowledge bases, real-time analytics, and AI-driven queries that align with Neo4j’s graph data platform capabilities.

(Table: Key touchpoints between Dinis Cruz and Neo4j, 2017–2025.)

This timeline demonstrates an ongoing engagement where Dinis not only uses Neo4j in innovative ways, but also consistently **brings new ideas to the Neo4j community**. From early adoption and public

praise of Neo4j’s capabilities, to organizing joint discussions with Neo4j’s team about future use cases, Dinis’s history reflects a mutually beneficial relationship that this proposal aims to elevate to the next level.

Current Work and Innovations by Dinis

Dinis’s current R&D efforts are highly relevant to Neo4j’s strategic directions, particularly in the areas of knowledge graphs, cloud deployment models, and AI integration. Below we detail his major projects and innovations:

Serverless Semantic Knowledge Graphs & MGraph-AI

One of Dinis Cruz’s foremost R&D themes today is the concept of **Serverless Semantic Knowledge Graphs**. This involves creating and manipulating rich knowledge graphs (with semantic context and provenance) in a serverless computing environment. The motivation is to combine the expressiveness of graphs with the scalability and cost-efficiency of serverless architectures.

To enable this, Dinis developed **MGraph-AI**, a new open-source graph database engine. Introduced in January 2025, MGraph-AI is described as a *“memory-first graph database, specifically designed for AI, semantic web, and serverless applications”* ³. Key characteristics of MGraph-AI include:

- **In-Memory Performance with JSON Persistence:** Graph data is kept in memory for speed, while being persisted as JSON to file storage for durability. This yields performance akin to an in-memory DB with the convenience of simple file-based snapshots ¹⁵.
- **Serverless Deployment Ready:** The design prioritizes use in ephemeral environments (e.g. cloud functions). MGraph-AI can spin up quickly and incur zero cost when not in use, unlike traditional graph databases that are “always on” ¹⁶ ¹⁷. This aligns with modern trends where applications need to scale out on-demand without long-running infrastructure.
- **Optimized for GenAI and Semantic Web:** The database supports use cases like LLM-powered knowledge retrieval and semantic search. For example, it’s built to handle **Graph-based RAG (Retrieval-Augmented Generation)** scenarios, where an LLM queries a knowledge graph for context. It also easily accommodates schema-free, evolving data (ontologies, taxonomies) typical of semantic web applications ¹⁸.
- **Type-Safe and Versionable:** MGraph-AI is implemented with type-safe data structures (leveraging Dinis’s earlier OSBot-Utils library) to catch errors early. It also stores graphs in a way that integrates with version control, enabling diffs and historical tracking of graph changes ¹⁹ – a feature valuable for collaborative knowledge base curation.

Relevance to Neo4j: MGraph-AI was born from attempting to use Neo4j in cutting-edge scenarios where conventional databases struggled. As Dinis noted, existing graph databases were *“too heavy or complex to deploy”* for his needs, particularly lacking easy serverless operability ¹⁶. By collaborating, Neo4j could gain insight from MGraph-AI’s approach. For instance, Neo4j’s cloud offering might incorporate a more lightweight spin-up option or file-based snapshotting for certain workloads. Additionally, the **semantic and AI-oriented features** of MGraph-AI highlight use cases (Graph + AI integration) that Neo4j is also exploring – providing a chance to join forces on reference implementations (discussed in Collaboration Opportunities section). Dinis’s work effectively complements Neo4j’s own Graph Data Science and vector search capabilities, pointing to a fusion of **knowledge graphs with AI** that could be very powerful.

MyFeeds.ai – A Case Study in LLM-Powered Knowledge Graphs

⁴ *Figure: MyFeeds.ai generates personalised news feeds by transforming raw articles into a knowledge graph and filtering them for each user persona.*

One of the tangible outputs of Dinis's recent research is **MyFeeds.ai**, an online MVP demonstrating how **semantic knowledge graphs and LLMs** can be combined to deliver personalized information feeds. MyFeeds.ai addresses a common challenge: information overload in cybersecurity news, and the need to tailor content to specific audiences (like a CEO vs a CISO).

At its core, MyFeeds uses an automated pipeline to turn unstructured text (news articles) into a structured **knowledge graph**:

- The system pulls articles from RSS feeds (initially sources like *The Hacker News*). Each article is processed by an LLM which performs **entity and relationship extraction**, effectively converting the article into a mini knowledge graph data structure ⁵. The graph nodes represent key entities (people, organizations, technologies, etc.) and edges represent semantic relationships (e.g. "Company X suffers Attack Y" linking an organization to an incident).
- These article graphs are enriched with provenance metadata, so every fact in the graph ties back to a source article and even the specific snippet. This addresses trust and traceability – a crucial aspect when using AI-generated data.
- Separately, MyFeeds creates "**persona graphs**" for each type of reader (e.g. a *CTO persona* might have a graph emphasizing cloud, DevOps, and business continuity topics). This is another LLM-generated knowledge graph that encodes what is important to that persona ²⁰.
- The article graphs and persona graphs are then compared (via graph similarity or embeddings) to select which news pieces best match each persona's interests. The output is a curated feed for the CTO, CISO, Board Member, etc., with each recommendation accompanied by *why* it was chosen (the graph relationships provide the explanation).

This entire workflow is implemented using Dinis's serverless graph tools (including MGraph-DB). The knowledge graphs exist as transient JSON-based graph structures that can be queried on the fly. Yet, the approach could be equally applied on a Neo4j platform, using Cypher and Neo4j's semantic search capabilities to store and query the extracted graphs.

Insight for Neo4j: MyFeeds.ai showcases an **innovative Graph+AI pipeline** that aligns with Neo4j's message of graph-fueled analytics. Neo4j executives and R&D directors can view this as a prototype for how enterprises might consume information in the future – by building knowledge graphs from text and using them for intelligent filtering/recommendations. Collaboration here could involve:

- Porting or re-implementing MyFeeds' pipeline using **Neo4j Graph Database** as the persistent store for the knowledge graphs, demonstrating Neo4j's ability to handle high-volume, dynamic graph data extracted by LLMs.
- Leveraging Neo4j's upcoming features (such as vector indexes for similarity search on text embeddings) in the persona matching step, to improve how articles are matched to user profiles.
- Showcasing Neo4j's **graph visualization** for the provenance graphs. For example, Neo4j Bloom or browser could present to users an interactive explanation graph of "why this article was recommended," adding transparency to AI – a very attractive feature for enterprise trust.

By working with Dinis on MyFeeds, Neo4j could produce a compelling case study (and perhaps a reference architecture) for **Personalized Threat Intelligence feeds**, useful for marketing to security

operations centers or any executives tracking cyber news. It's a concrete demonstration of Neo4j's relevance in the GenAI era, turning unstructured data into a navigable knowledge network.

The Cyber Boardroom: Vision for AI-Assisted Governance

Dinis's flagship venture, **The Cyber Boardroom**, encapsulates his business vision for applying graph and AI technologies to a pressing industry need: bridging the communication gap between cybersecurity leaders and corporate boards. This is a domain where Neo4j's technology could play a foundational role, and thus it deserves special attention as a collaboration opportunity.

Concept: The Cyber Boardroom is envisioned as a **GenAI-powered platform** that distills complex cybersecurity information into tailored insights for board members. With less than 9% of directors having tech/security backgrounds, boards struggle with oversight while CISOs struggle to explain cyber risk in business terms ²¹. The platform aims to solve this by creating an intelligent intermediary: essentially a set of AI "personas" that translate between the technical and the strategic. According to Dinis's business plan, *"The Cyber Boardroom ... bridges the knowledge gap between cybersecurity experts and corporate boardrooms. Built on an open-source foundation with modern serverless architecture, it leverages advanced AI personas (Athena, Minerva, and Odin) to enable clear, two-way communication through tailored, easy-to-understand cybersecurity insights."* ¹³.

Under the hood, this requires a robust **knowledge graph** of an organization's cybersecurity posture and broader threat landscape: - The knowledge graph would encompass assets, threats, controls, incidents, and compliance obligations – linked in a way that queries can be answered in plain language (e.g. "How exposed are we to ransomware threats in our supply chain?"). - AI personas (like "Athena" for the Board and "Odin" for the CISO, as referenced in the plan) would interface with this graph. Essentially, an LLM would use the graph data to generate narratives or answer questions, ensuring that responses are both accurate (grounded in the graph's facts) and audience-appropriate. - By design, the solution emphasizes **traceability** (each statement to the board can be traced to data and risk models in the graph) and **determinism** (unlike a black-box AI, it doesn't invent facts – it pulls from the curated graph).

Dinis has progressed this project to the point of a detailed business plan and a call for investment, seeking **£250k seed funding** to accelerate development and go-to-market ¹⁴. The emphasis on an open-source core and serverless deployment suggests he intends the platform to be flexible and widely integrable – traits that align well with using Neo4j as a backbone.

Potential Role of Neo4j: Neo4j's graph database is an excellent candidate to serve as the core knowledge graph repository for The Cyber Boardroom. Neo4j's strengths (scalability, ability to model complex relationships, ACID transactions for data integrity, and a rich query language in Cypher) are well-suited to capturing a dynamically evolving risk and governance knowledge base. Furthermore, Neo4j has existing **graph algorithms** and **graph data science** libraries that could enhance the platform (e.g. running centrality algorithms to find key risk concentrations, or similarity algorithms to map emerging threats to company assets).

By collaborating with Dinis on The Cyber Boardroom, Neo4j stands to benefit in several ways:

- **Strategic Market Penetration:** The platform targets corporate boards and executives – an audience Neo4j is keen to engage (as potential customers of Neo4j's technology via their organizations). A success here would demonstrate graph databases' value at the highest levels of decision-making, not just in technical departments.

- **Product Feedback and Enhancement:** Dinis's requirements (e.g. needing to run on serverless infrastructure, or handle certain on-the-fly queries) could inform Neo4j's product roadmap. For instance, if The Cyber Boardroom needs a way to do easy **what-if analyses** on the graph or quick spin-up of a graph instance for a single meeting's briefing, Neo4j's team could explore features to accommodate that (making Neo4j more flexible for other customers too).
- **Thought Leadership and Co-Marketing:** A joint case study or whitepaper on how Neo4j empowers The Cyber Boardroom's solution would be powerful content. It intersects hot topics – cybersecurity, board governance, AI – with Neo4j's technology. Neo4j could showcase this as an innovative use of their product in the GenAI age, attracting attention in both the data and security communities.

In summary, The Cyber Boardroom represents a **high-level, enterprise-focused application** of graph technology. Neo4j's collaboration here could yield a flagship success story, illustrating how graphs can tackle governance challenges. Dinis's vision aligns well with Neo4j's mission to make sense of data relationships, and by working together, both parties can accelerate the realization of this vision.

Ongoing Open-Source Projects and Contributions

Beyond the marquee projects above, Dinis remains actively involved in a number of open-source initiatives, many of which could tie into Neo4j's ecosystem. A few noteworthy mentions:

- **Project JSync (Jira Exporter & Sync System):** This is a tool Dinis designed to extract data from Atlassian Jira and synchronize it with external stores (like graph databases). As part of his exploration of Jira-as-Graph, he outlined a *"Project Plan: JIRA Exporter to S3 and Git"* which can serve as a pipeline to liberate issue data ²². Combined with Neo4j, JSync could enable organizations to continuously feed their project management data into Neo4j, turning Jira projects into living graphs (for risk management, dependencies mapping, etc.).
- **Project Lumos (Serverless JIRA-to-GraphDB Connector):** Building on the above, Project Lumos is conceptualized as a serverless connector that listens to Jira events and updates a **GraphDB** in real-time. The "GraphDB XYZ" in Dinis's documentation implies it could work with any graph database – making Neo4j an ideal candidate. Essentially, this would allow Neo4j to be seamlessly updated with the latest information from Jira, enabling use cases like real-time impact analysis of new software vulnerabilities or tracking development risks through a graph. Such a connector could be turned into a Neo4j integration offering, with Dinis's development effort paving the way.
- **OSBot and Automation Tools:** Dinis was a leader in the OWASP Security Bot project (OSBot), creating utilities (OSBot-Utils, OSBot-AWS, etc.) that automate security tasks. While not graph databases per se, these tools reflect Dinis's knack for automation and could complement Neo4j in scenarios like automated threat data ingestion. For example, OSBot scripts could feed Neo4j with vulnerability scan results or cloud configuration data, building a comprehensive security knowledge graph.
- **Research Publications:** On his Docs & Research site, Dinis frequently publishes deep dives that often involve graph concepts. Recent papers like "Advancing Threat Modeling with Semantic Knowledge Graphs" and "Semantic OWASP: Leveraging GenAI and Graphs to Customize and Scale Security Knowledge" (2025) indicate his forward-thinking approach to combining **open standards, knowledge graphs, and AI**. These writings not only influence the community but could also influence Neo4j's own R&D if leveraged. Neo4j might partner with Dinis to co-author research or develop PoCs based on these ideas, further cementing Neo4j's authority in new applications of graph tech.

In all these efforts, a common thread is that Dinis builds **bridges** – between data silos and graphs, between AI and structured knowledge, and between practitioners and tools. His open-source mindset

ensures the work is transparent and extensible. Neo4j can confidently collaborate knowing that solutions arising from Dinis's projects can integrate with or run on the Neo4j platform, benefiting both Neo4j's user community and the wider open-source ecosystem.

Previous Collaboration Ideas (Open Security Summit 2022)

It is worth revisiting the **Open Security Summit March 2022 session** ("Ideas for Graph DBs like Neo4j") as it encapsulates several collaboration ideas that Dinis and Neo4j participants identified together. During that session, a range of inventive uses for Neo4j in security and IT governance were discussed ¹². Some highlights included:

- **Graphs in Threat Modeling:** Using Neo4j to model threat scenarios, attack trees, and mitigations. This could help visualize how an attack can traverse an organization's assets and where controls exist (or are missing). Neo4j's ability to handle complex relationships is a natural fit, and Dinis proposed that formal threat models could be stored and queried as graphs (as opposed to static documents).
- **Cyber Threat Intelligence (CTI) Graphs:** Mapping cyber threat intel feeds (like STIX/TAXII data) into Neo4j for enrichment and analysis ²³. By connecting indicators of compromise, threat actors, tactics, etc., Neo4j can reveal hidden linkages. The session noted that there are community projects (e.g. GraphKer by Adamantios M. Berzovitis) already using Neo4j in this way, and Dinis has shown interest in building on those approaches.
- **Risk Management and Governance:** Representing risk registers, audit findings, and compliance controls as nodes and edges. An idea was that Neo4j could become a "single source of truth" for governance data. For example, linking a risk (node) to the assets it affects, the controls mitigating it, and the audit tests validating those controls forms a graph that answers complex questions ("Which risks are tied to these five systems and who owns them?"). Dinis has experience in governance processes and sees graph DBs as a superior way to handle this interconnected information ²⁴.
- **DevOps and Infrastructure Mapping:** The session touched on using Neo4j to map network topologies, cloud resources (like AWS IAM relationships), and even developer workflows ²⁵. One specific idea was using **GitHub repositories to store graph data**, effectively version-controlling the graph (something Dinis later implemented conceptually with JSON storage in MGraph). Another was visualizing infrastructure as code dependencies with Neo4j, aiding in impact analysis of changes.
- **Jira + Neo4j Fusion:** A particularly interesting Neo4j-specific topic raised was *comparing Neo4j with Jira as a graph store* ⁸. Dinis posited (and continues to explore) that Jira's issue-linking can form a primitive graph DB, but Neo4j offers far more powerful querying and flexibility. The collaboration idea here is to integrate the two: use Jira for what it's good at (workflow, user interface) and Neo4j for deep graph analytics. This reinforced the concept for a connector (now Project Lumos) to keep the two in sync, so companies can use Neo4j's superior graph queries on live Jira project data.
- **Visualization and UX:** Dinis and colleagues noted the importance of making graph data accessible to end users. Ideas included using lightweight JavaScript libraries to embed Neo4j visualizations in other platforms, generating **static site content from Neo4j** (e.g. using Hugo to publish dashboards from graph queries), and even treating **Neo4j instances as disposable "cattle"** rather than permanent "pets" ²⁶. The latter metaphor, drawn from cloud DevOps practices, suggests spinning up Neo4j instances for specific analysis tasks or tests and tearing them down – a practice that dovetails with Dinis's serverless direction.

These brainstorming points from 2022 illustrate that many **collaboration ideas have already been put on the table**. What remains is to execute and expand on them. The presence of Neo4j staff in that

discussion (e.g. Neo4j's Developer Relations and solution engineers) showed a mutual interest in these directions. This proposal aims to carry those ideas forward, with concrete next steps to realize their potential.

Proposed Collaboration Areas with Neo4j

Considering Dinis Cruz's expertise and Neo4j's strategic aims, several **concrete collaboration opportunities** emerge. These are designed to align Dinis's ongoing research with Neo4j's product roadmap and market goals, creating win-win outcomes:

1. Serverless Neo4j & Edge Deployments

Proposal: Jointly explore a **"Neo4j Serverless"** concept, leveraging Dinis's MGraph-AI insights. This could involve a lightweight Neo4j mode or toolkit for running Neo4j in ephemeral environments (e.g. AWS Lambda, Azure Functions, or as part of containerized pipelines). Dinis can contribute his experience in achieving fast startup, memory-optimized graph operations, and JSON-based state storage ¹⁸.

Neo4j Benefit: Attract new developers who need graph capabilities in serverless apps, and reduce friction for using Neo4j in CI/CD, data processing jobs, or edge computing scenarios. This aligns with Neo4j's push into cloud-native deployment and could expand usage in microservice architectures.

Collaboration Activities: Form a small R&D task force (Dinis and Neo4j engineers) to prototype a serverless-friendly Neo4j variant or an adapter that converts between in-memory JSON graphs and Neo4j. Document best practices for "Neo4j as cattle" deployments (dynamic spin-up/down) ²⁶. Possibly integrate some MGraph-AI features into Neo4j Aura (Neo4j's cloud service) for an offering that pauses when not in use, saving cost.

2. Knowledge Graphs for GenAI (Graph+LLM Integration)

Proposal: Develop a reference architecture and toolkit for **Graph-augmented LLM applications** using Neo4j. Dinis's MyFeeds.ai is a perfect case study – converting text to graph and using the graph for LLM context ⁵. The collaboration would generalize this: create Neo4j procedures or APIs that assist in storing LLM-extracted knowledge and retrieving it during LLM queries (Graph-RAG pattern).

Neo4j Benefit: Strengthen Neo4j's position in the hot area of AI/LLM integration. By showcasing Neo4j as the knowledge graph that grounds AI in facts (preventing hallucinations through provenance ⁴), Neo4j can appeal to enterprises looking to tame AI with their own data. This also promotes Neo4j's recent features like vector indexing and Graph Data Science in AI use cases.

Collaboration Activities: Co-author a whitepaper and sample code on building an AI-powered knowledge graph workflow (possibly "Neo4j + OpenAI for News Intelligence" as a concrete example). Host a webinar or Neo4j DevZone article featuring Dinis demonstrating how an LLM (ChatGPT or similar) can interact with a Neo4j knowledge graph in a Q&A format. Long-term, explore productizing a Neo4j plugin that interfaces with LLMs – e.g. a Cypher function that calls out to an LLM for entity extraction, or vice versa, an LLM that knows how to query Neo4j via Cypher.

3. Jira-Neo4j Integration for DevSecOps

Proposal: Build an official **Jira-to-Neo4j connector** in partnership with Dinis, inspired by his Project JSync/Lumos. This tool would synchronize Jira issues (and their relationships) into Neo4j in near real-

time, enabling graph-based analytics on project and risk data. The connector could be an open-source Neo4j Labs project or even an offering on the Atlassian Marketplace.

Neo4j Benefit: Tapping into Jira's huge user base, Neo4j could become the go-to graph database for visualizing and analyzing work items, dependencies, and risks in software projects. This drives Neo4j adoption in the DevOps and project management community. It also addresses a known gap – while Jira contains graph-like data, it's hard to query globally; Neo4j can fill that gap ²⁷ ⁸ .

Collaboration Activities: Neo4j provides engineering support and distribution channels, Dinis provides the domain design and prototype (he has already outlined how to map Jira data to a graph schema). Together, define use cases: e.g., "Dependency Impact Graph" (which Jira tasks or components are most interconnected), "Risk Heatmap" (link Jira issues to risk register in Neo4j), "Organizational Network" (based on who is reporting to whom in tasks, etc.). Deliver a connector with easy setup, possibly using Atlassian webhooks and Neo4j's REST/GraphQL API. Market this via a case study (e.g. "How Glasswall used Neo4j with Jira to achieve SOC2 compliance" – a plausible story given Dinis's background in automating SOC2 with Jira).

4. Cybersecurity Knowledge Graph Solutions

Proposal: Co-create solution frameworks for **cybersecurity analytics using Neo4j**. This would package Neo4j use-cases that Dinis has been championing – such as threat intelligence graphs, attack path analysis, security controls mapping, and incident response knowledge graphs – into reference solutions or starter kits. Essentially, turn ideas from the 2022 summit ²⁸ ⁷ into tangible modules.

Neo4j Benefit: Establish Neo4j as the premier graph platform in the cybersecurity sector. Many security teams are not graph experts; providing pre-built schemas and queries for common security problems lowers the barrier to entry. Neo4j can thus expand its customer base in a domain that values connected data (attacks often traverse systems, which is inherently graphy). This could also drive usage of Neo4j Graph Data Science library (for detecting attack patterns, scoring risk propagation, etc.).

Collaboration Activities: Identify 2–3 high-impact security use cases. For example: **Threat Detection Graph** (ingest SIEM logs or vulnerability scan data into Neo4j to find connections indicating threats), **Identity & Access Graph** (map IAM relationships to find privilege escalation paths), **Compliance Graph** (map standards to controls to evidence). Dinis provides the domain models and sample data from his experience; Neo4j provides technical refinement. Create content (blog series, conference talks, or even an "Neo4j for Cybersecurity" mini-course) with Dinis as a co-presenter. This not only yields technical assets but also leverages Dinis's credibility in the security community to boost Neo4j's profile.

5. The Cyber Boardroom Pilot with Neo4j

Proposal: Form a strategic partnership around **The Cyber Boardroom** platform. Concretely, Neo4j could become an early technology partner (or even investor) to ensure that Neo4j's database is the engine powering The Cyber Boardroom's knowledge graph. This could start as a pilot deployment for one or two joint clients (e.g. a friendly board or CISO willing to try the platform with their data, under guidance from Dinis and Neo4j).

Neo4j Benefit: If The Cyber Boardroom gains traction, it will drive enterprise adoption of Neo4j in a top-down manner (boards asking their organizations to implement the solution, which inherently uses Neo4j). Even aside from direct financial returns, Neo4j stands to learn a great deal about the requirements of high-level decision-support graphs – informing product improvements for analytics, security (e.g. fine-grained access control for board-level data), and natural language query interfaces.

There's also public relations value in being associated with an initiative that promises to improve corporate cybersecurity oversight (a timely issue for regulators and investors).

Collaboration Activities: Neo4j could assign an “*Innovation Partner*” liaison to work with Dinis as he builds out the MVP. This person/team would help optimize the data model in Neo4j, ensure performance at scale, and integrate Neo4j Aura or Bloom as needed. In exchange, Neo4j gets a showcase deployment. Additionally, consider funding or resource support – the requested seed investment (£250k) is relatively small; Neo4j might participate in that round or provide cloud credits and technical sponsorship, signaling confidence in the venture. Joint press releases or conference presentations (e.g. at Neo4j's GraphConnect or a major security conference) can announce the partnership and demonstrate how Neo4j technology underpins this innovative platform.

6. Community and Evangelism Initiatives

Proposal: Engage Dinis as a **guest evangelist/advisor** in Neo4j's community programs. For example, Dinis could host Neo4j-sponsored webinars on graph security, contribute articles to Neo4j's blog (sharing his experiences), or speak on behalf of Neo4j at industry events (like RSA Conference, OWASP Global AppSec, etc.) about graph-based approaches.

Neo4j Benefit: Dinis offers a rare combination of credibility in both the security realm and the developer community. His endorsement and thought leadership can attract new users from sectors that Neo4j wants to grow (cybersecurity, enterprise governance, etc.). By aligning with his respected voice, Neo4j strengthens its position as not just a database vendor, but a solution provider for pressing problems. It also helps humanize the Neo4j brand – through Dinis's authentic narratives of solving real problems with graphs.

Collaboration Activities: Include Dinis in Neo4j's **Innovation Advisory Board** or a similar panel if one exists – giving him direct input channels to Neo4j's product strategy. Arrange a series of talks or workshops: e.g., “Graph Technology in Cybersecurity” mini-summit (possibly as part of Open Security Summit events) with Neo4j's sponsorship. Support Dinis's open-source projects under the Neo4j banner where appropriate (for instance, adding them to the Neo4j Labs incubator if they align). Such community collaboration would be relatively low-cost but high-impact in spreading Neo4j knowledge and generating goodwill.

Each of the above collaboration proposals is **actionable and aligned with Neo4j's interests**. They range from technical R&D (serverless, LLM integration) to go-to-market strategy (joint solutions, industry evangelism). Neo4j can opt to pursue one or multiple tracks, phased as immediate wins vs. longer-term projects. What is common to all is that Dinis Cruz's involvement and intellectual capital would accelerate Neo4j's progress in the area, while Neo4j's platform and resources would amplify the impact of Dinis's work.

Strategic Alignment with Neo4j's Vision

Neo4j's vision, as a leader in graph databases, has evolved to encompass not just storing data, but enabling **knowledge discovery, intelligent applications, and broad enterprise adoption** of graph technology. Collaborating with Dinis Cruz directly supports this vision in several ways:

- **Innovation at the Cutting Edge:** Dinis operates where multiple frontiers meet – graphs, AI, cybersecurity. By integrating his breakthroughs (like semantic graphs with LLMs) into Neo4j's

ecosystem, Neo4j stays at the forefront of technological innovation. This ensures the Neo4j Graph Data Platform remains relevant as AI-driven applications proliferate.

- **Use-Case Diversification:** Neo4j has traditionally seen strong adoption in areas like fraud detection, recommendation engines, and network IT ops. The collaboration highlights use-cases in *cybersecurity and governance*, which are growth areas. Expanding into these domains with credible solutions (backed by Dinis's experience) can open new market segments for Neo4j.
- **Cloud and Deployment Model Leadership:** The industry is trending toward cloud-native and serverless architectures. Dinis's work on serverless graph usage gives Neo4j an opportunity to lead in this space, ensuring that Neo4j is not seen as a legacy, heavyweight system but as a flexible component in modern stacks. This complements Neo4j's existing cloud services (AuraDB) and could inspire new offerings or improvements.
- **Community and Open-Source Goodwill:** Dinis's open-source contributions and community events align with Neo4j's own developer community focus. Embracing his projects under the Neo4j umbrella (or co-marketing them) sends a message that Neo4j supports open innovation. This can attract developers who appreciate transparency and collaborative development.
- **Public Relations and Thought Leadership:** There is an increasing public spotlight on cybersecurity (boardrooms and governments are paying attention). A partnership that results in, say, a high-profile story of "Neo4j enabling the next-gen cybersecurity oversight platform" could garner significant positive PR. It positions Neo4j as a solution to real, timely problems, not just a database vendor. Executives at Neo4j can point to such collaborations as evidence of the company's commitment to solving big-picture challenges.

In essence, the proposed collaboration is **strategically synergistic**. Dinis Cruz's pursuits underscore many of the directions Neo4j is heading: more cloud-native, more integrated with AI, more domain-specific solutions. By formally aligning efforts, Neo4j can accelerate progress on these fronts while Dinis gains a robust platform and backing for his initiatives. This synergy exemplifies the principle that the whole can be greater than the sum of its parts.

Conclusion

Dinis Cruz's track record with Neo4j – from being a power user and community advocate to pioneering new graph applications – speaks to the **natural alignment** between his work and Neo4j's mission. This proposal has outlined how, in concrete terms, a closer collaboration can yield mutual benefits: Neo4j extends its technological and market lead, and Dinis's ideas reach their full potential with the support of an enterprise-grade platform and team.

Next Steps: It is recommended that Neo4j's executive team designate a point person to engage with Dinis in detailing a collaboration plan. An initial meeting or workshop can be organized (perhaps even as a session at an upcoming Neo4j event or Open Security Summit) to prioritize the collaboration areas listed. From there, a roadmap with joint milestones can be developed – for example, delivering a Neo4j+MyFeeds demo within 3 months, a Jira connector by Q4, and a Cyber Boardroom pilot in the new year.

With Neo4j's resources and Dinis Cruz's innovation combined, the company can drive forward a new generation of **graph-powered solutions**. This partnership proposal is not just about supporting one researcher's work; it is a strategic investment in advancing what graphs can do in an era increasingly defined by data relationships and intelligent automation. By seizing this opportunity, Neo4j can reinforce its position as the graph platform of choice for solving the most challenging and exciting problems in technology today.

References:

1. Cruz, D. (2022). *Ideas for Graph DBs like Neo4j – Open Security Summit (Session Notes)* ⁷ ⁸ – Discussion topics highlighting use cases such as serverless graph databases and Neo4j's role alongside Jira.
2. Neo4j Community Blog (2018). *"This Week in Neo4j"* ⁶ – Feature of Dinis Cruz's work importing GDPR data from Jira into Neo4j, demonstrating early innovation in graph usage.
3. Cruz, D. (2025). *"Introducing MGraph-AI – A Memory-First Graph DB for GenAI and Serverless Apps"* ³ ¹⁸ – LinkedIn article by Dinis Cruz explaining the motivations and features of the MGraph-AI open-source project.
4. Cruz, D. (2025). *MyFeeds.ai MVP – "Establishing Provenance and Deterministic Behaviour in an LLM-Powered News Feed"* ⁴ ⁵ – Technical post detailing how MyFeeds uses semantic knowledge graphs to personalise content with full provenance.
5. Cruz, D. (2023). *The Cyber Boardroom – Investor Business Plan* ¹³ ¹⁴ – LinkedIn post outlining the vision, architecture (open-source, serverless, AI personas), and funding strategy for The Cyber Boardroom platform.
6. Open Security Summit (Mar 2022). *Session: Ideas for Graph DBs like Neo4j* ¹² ²³ – Summit page confirming the focus on Neo4j use cases in security, co-led by Dinis Cruz with Neo4j team participation.
7. Cruz, D. – OWASP London (2019). *"Creating a Graph-Based Security Organisation"* ¹⁰ – Presentation (originally DevSecCon 2017) by Dinis demonstrating graph-based approaches to security, reflecting his long-standing advocacy for Neo4j-like solutions.
8. Dinis Cruz – Bio/Profile (Open Security Summit) ¹ ⁹ – Background on Dinis's expertise (20+ years in cybersecurity and software) and industry recognition.

¹ ⁹ Dinis Cruz

<https://open-security-summit.org/participant/glasswall/dinis-cruz/>

² Building Semantic Knowledge Graphs with LLMs: Inside MyFeeds.ai's Multi-Phase Architecture

<https://www.linkedin.com/pulse/building-semantic-knowledge-graphs-llms-inside-myfeedsais-dinis-cruz-jub5e>

³ ¹⁵ ¹⁶ ¹⁷ ¹⁸ ¹⁹ Introducing: MGraph-AI - A Memory-First Graph Database for GenAI and Serverless Apps

<https://www.linkedin.com/pulse/introducing-mgraph-ai-memory-first-graph-database-genai-dinis-cruz-wxmde>

⁴ ⁵ ²⁰ Establishing Provenance and Deterministic Behaviour in an LLM-Powered News Feed (first MyFeeds.ai MVP)

<https://mvp.myfeeds.ai/establishing-provenance-and-deterministic-behaviour-in-an-llm-powered-news-feed-first-myfeeds-mvp/>

⁶ nlp Archives - Neo4j Graph Data Platform

https://graph1283.rssing.com/chan-60420690/all_p1.html

⁷ ⁸ ¹² ²³ ²⁴ ²⁵ ²⁶ ²⁸ Ideas for Graph DBs like Neo4j

<https://open-security-summit.org/sessions/2022/mini-summits/mar/governance/ideas-for-graph-dbs-like-neo4j/>

¹⁰ Creating a graph based security organisation - Apr 2019 (OWASP ...)

<https://www.slideshare.net/slideshow/creating-a-graph-based-security-organisation-apr-2019-owasp-london-chapter-meeting/140239010>

¹¹ ²² ²⁷ Jira as Graph DB | Dinis Cruz | 10 comments

https://www.linkedin.com/posts/diniscruz_jira-as-graph-db-activity-7293099257547354112-z4xs

13 14 21 the Cyber Boardroom - Business Plan | Dinis Cruz

https://www.linkedin.com/posts/diniscruz_the-cyber-boardroom-business-plan-activity-7296167423597039617-pC0c